BOILEAU GUILLAUME

Systems & RAMS Engineer | Safety, Reliability, IVVQ, Risk Analysis & Space Systems

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EXPERIENCE

Software Engineer – System Integration, Operational Reliability & Production Diagnostics

VEV Platform Services France

2025 – 2026 | Valbonne, France

- Developed and maintained web and backend services for an EV charging CPMS platform using **TypeScript, Angular and Node.js**.
- Contributed to software integration activities involving operational charging infrastructure, hardware interfaces and field equipment constraints.
- Analysed hardware/software interactions, operational data exchanges and service behaviours in production.
- Investigated incidents, abnormal behaviours, communication issues and data inconsistencies affecting service continuity.
- Supported verification, correction and monitoring of operational data flows, including FTP-based exchanges.
- Contributed to issue diagnosis, prioritization and follow-up with technical stakeholders in an industrial operational context.
- Strengthened experience with reliability, maintainability, service robustness and production constraints for software systems connected to physical infrastructure.

Senior R&D Engineer – Space Systems, Validation, Risk Analysis & Technical Coordination

Laboratoire Lagrange, Observatoire de la Côte d'Azur, CNES

2023 – 2025 | Nice, France

- Contributed to engineering and analysis activities for the **LISA ESA/NASA space mission**, in an international environment involving complex mission-level requirements, scientific performance constraints and high technical rigor.
- Supported activities aligned with **V-cycle engineering logic**: requirements interpretation, validation planning, consistency checks, result verification, anomaly analysis and technical evidence consolidation.
- Developed CATpyLISA, a Python platform for large-scale model integration, comparison, analysis and validation workflows.
- Structured validation workflows to compare outputs, verify expected behaviours, detect deviations, analyse non-conformities and consolidate technical evidence.
- Worked at the interface between scientific requirements, software development, data processing, modelling assumptions and mission-level performance objectives.
- Identified inconsistencies, limitations, uncertainty sources and critical behaviours in complex software and analysis chains.
- Produced risk-oriented technical analyses by evaluating the impact of modelling choices, numerical behaviours and data-processing assumptions on final results.
- Supported traceability of configurations, parameters, assumptions, validation results and technical conclusions.
- Contributed to technical review preparation through documentation of validation procedures, limitations, assumptions and interpretation of results.
- Coordinated technical activities involving contributors, research engineers, technicians and interns.
- Supervised Master's thesis interns on software, data analysis, validation and modelling tasks.
- Operated in a collaborative international framework requiring technical English, structured reporting, versioned workflows and rigorous documentation.

Data Scientist – Instrument Performance, Noise Modelling & Failure Diagnostics

Universiteit Antwerpen, Belgium

2021 – 2023 | Antwerp, Belgium

- Analysed instrumental and environmental effects impacting the sensitivity, stability and performance of precision measurement systems.
- Developed Python-based analysis pipelines for noise source identification, uncertainty estimation, Bayesian inference and statistical modelling.
- Investigated unexpected behaviours, degradation patterns and correlations between environmental conditions and measurement performance.
- Supported technical interpretation of system limitations by identifying dominant contributors, weak points and possible mitigation directions.
- Produced reproducible and traceable technical analyses in a collaborative international environment.
- Contributed to studies connected to gravitational-wave detector systems, where performance, noise control, robustness and system-level sensitivity are critical.
- Communicated complex technical results to scientific and engineering stakeholders through structured reporting and collaborative review.

SUMMARY AND STRENGTHS

Systems and RAMS-oriented engineer with strong experience in complex system validation, risk analysis, reliability-oriented diagnostics and performance assessment in space and industrial software environments. Background developed through major international collaborations including **LISA**, **LIGO–Virgo–KAGRA** and **Einstein Telescope**, complemented by recent industrial experience involving hardware/software interfaces, operational reliability and production issue analysis.

Experienced in **requirements-oriented validation**, **IVVQ support**, **risk identification**, **failure diagnostics**, **anomaly investigation**, **root-cause-oriented analysis**, **traceability**, **technical documentation**, **Python automation**, **software integration** and **cross-functional technical coordination**.

Able to analyse complex systems, identify critical behaviours, consolidate technical evidence, document assumptions and support design, mitigation or conformity decisions in high-constraint environments.

Relevant for **RAMS**, **Safety**, **Reliability**, **Availability**, **Maintainability**, **system verification**, **risk mitigation**, **technical reviews**, **space systems engineering** and **complex system validation**.

FIT FOR THALES RAMS / SAFETY

- Experience with complex space systems, mission-level performance constraints and precision-instrument environments requiring rigorous validation and technical evidence.
- Strong ability to identify failure scenarios, uncertainty sources, degraded behaviours, weak points and residual risks in complex software/data/system chains.
- Background in verification, validation, traceability, result consolidation, configuration follow-up and technical documentation.
- Experience in anomaly investigation, root-cause-oriented analysis and diagnostic reporting in both research and industrial software contexts.
- Familiar with system-level constraints, reliability of analysis chains, maintainability of technical workflows and robustness of operational services.
- Able to work at the interface between requirements, modelling, software, data, system performance and operational constraints.
- Relevant background for RAMS activities involving reliability, availability, maintainability, safety, risk analysis, conformity evidence and technical reviews.
- Comfortable in transnational and multidisciplinary environments requiring technical English, structured reporting, stakeholder alignment and execution rigor.

TEAM MANAGEMENT

- Supervision of **3 Master's thesis interns (M2)** during engineering, modelling and software-development phases.
- Coordination of technical activities involving **research engineers and technicians** on a **data processing pipeline** for the **LISA space mission**.
- Experience in organizing tasks, supporting contributors, reviewing outputs and maintaining alignment across collaborative technical developments.

LANGUAGES

English	●●●●○
Spanish	●●○○○

Junior R&D Engineer – Signal Analysis, Simulation & System Performance Assessment

ARTEMIS, Observatoire de la Côte d'Azur

2018 – 2021 Nice, France

- Conducted research and engineering-oriented analysis for the preparation of gravitational-wave space missions, with a strong focus on **LISA**.
- Developed methods for the separation of weak overlapping signals in complex low-SNR datasets.
- Built simulation and analysis workflows to evaluate system performance under realistic noise, foreground and uncertainty conditions.
- Compared simulated outputs, model behaviours and analysis results to assess consistency, robustness and performance limitations.
- Investigated noise sources and correlations through diagnostic analyses relevant to precision instruments.
- Produced peer-reviewed results contributing to the understanding of performance limits for future gravitational-wave space missions.
- Developed strong expertise in modelling, uncertainty analysis, signal processing, technical validation and scientific software workflows.

PROFESSIONAL TRAINING

Supervised Machine Learning

DeepLearning.AI – Andrew Ng

2020 Coursera

Recherche reproductible : principes méthodologiques pour une science transparente

FUN MOOC – Inria

2025 France Université Numérique

- Training focused on reproducible research practices, computational documentation, data management, version control and transparent analysis workflows.
- Practical exposure to **Git/GitLab**, **Jupyter notebooks**, structured documentation, data handling and reproducible software environments.

Continuous Professional Training – Software, Data, AI and Systems

OpenClassrooms

2024 – 2026 28 completed courses

- Completed courses covering **Python**, **JavaScript**, **TypeScript**, **Angular**, **Node.js**, **Django**, **REST APIs**, **SQL**, **Machine Learning**, **AI**, **Linux**, **C/C++**, web development and software engineering fundamentals.
- Strengthened practical skills in full-stack development, data analysis, algorithmic thinking, database querying, Linux administration and modern software tooling.

EDUCATION

PhD in Physics

Université Côte d'Azur

2018 – 2021 Nice, France

Selective Graduate Program in Fundamental Physics

Université Paris-Saclay

2016 – 2018 Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

2017 – 2018 Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

2015 – 2016 Orsay, France

PROJECTS

Co-author and full member of the LISA Consortium

ESA/NASA Space Mission – Large-scale International Collaboration

2018 – now

Contribution to software, analysis and validation workflows for the LISA space mission within an international engineering and scientific collaboration involving ESA, NASA and multiple research institutes.
Role: Contribution to software workflows, technical coordination, complex signal-processing activities, model integration, validation campaigns, performance assessment and collaborative engineering tasks.
RAMS relevance: Work involving traceability, consistency checks, performance limitations, uncertainty analysis, model comparison, validation evidence and identification of critical behaviours in complex mission-related workflows.
Project relevance: Work at the interface between software development, scientific requirements, data processing, validation workflows, technical contributors and international coordination.
Program scale: Major international space mission with an estimated budget of approximately 2 billion EUR over 10 years.

Co-author and full member of Einstein Telescope and LIGO–Virgo–KAGRA Collaborations

International Collaborations

2021 – now

Contribution to collaborative developments in data analysis, signal characterization, software methods and technical investigations within the LIGO–Virgo–KAGRA and Einstein Telescope collaborations.
Role: Development of analysis tools, contribution to technical activities and support to complex software workflows in high-standard collaborative environments.
System impact: Contributed to studies on environmental and instrumental noise sources, helping identify fundamental design constraints and performance limitations for next-generation gravitational-wave observatories.
Environment: Large-scale international collaborations involving strong technical requirements, structured methods, technical review, versioned software workflows and coordinated contributions.

STRENGTHS

RAMS, Safety & risk analysis

RAMS | Safety | Reliability | Availability | Maintainability | Risk Analysis | Risk Identification | Failure Diagnostics | Failure Scenario Analysis

Critical Behaviour Analysis | Residual Risk Awareness | Risk Mitigation Support | System Robustness | Performance Consolidation | Uncertainty Analysis | Conformity Evidence | Safety-Oriented Engineering

Systems & validation engineering

Systems Engineering | Space Systems | Satellite Mission Context | Mission-Level Performance | Requirements-Oriented Validation | Verification & Validation | System Validation | Validation Campaigns | Test Strategy Support | Software IVVQ | Software Integration | Integration Testing | Requirements Traceability | Anomaly Analysis | Root Cause Analysis | Technical Diagnostics | Issue Resolution | Hardware / Software Interfaces | Operational Interfaces

Technical coordination

Technical Project Coordination | Work Package Structuring | Task Follow-Up | Contributor Alignment | Technical Roadmap Support | Delivery Support | Technical Reporting | Stakeholder Communication | Cross-Functional Coordination | Team Coordination | Issue Follow-Up | Decision Support | Documentation Quality | Structured Execution

Technical skills

Python | Shell Scripting | Bash | Linux | Git | GitLab | Docker | CI/CD | Test Automation Scripting | SQL | TypeScript | Angular | Node.js | C | Matlab | Pandas | REST API | FTP | Issue Tracking | Jira | Confluence | OpenSearch | Operational Monitoring | Technical Reporting | Condor | Slurm | PyTorch | MongoDB | Mathematica

Domain knowledge

Space Systems | Satellite Systems | ESA/NASA Collaboration | Transnational Environment | Scientific Payload Analysis | Precision Instrumentation | Space Mission Data Processing | Large-Scale International Projects | Critical Systems Awareness | Industrial Software | Technical English

Soft skills

Autonomy | Analytical Thinking | Execution Rigor | Problem Solving | Adaptability | Responsibility | Attention to Detail | Collaboration | Structured Communication | Customer Awareness | Technical Interface | Leadership Potential